

Pure Substance

1. A two-phase liquid vapour mixture of H_2O is at 30 bar. If on heating at fixed volume, critical point is attained. Determine the quality at the initial state. [PU 2010 Fall]
2. A vessel having a volume of $0.4m^3$ contains 2 kg of liquid water and water vapor mixture in equilibrium at a pressure of 0.6 MPa. Calculate (a) the volume and mass of liquid. (b) the volume and mass of vapor. [PU 2009 Spring]
3. 5 kg of H_2O are contained in a closed rigid tank at an initial pressure of 2000kPa and a quality of 50%. Heat transfer occurs until the tank contains only saturated vapour. Determine the volume of the tank, in m^3 , and the final pressure, in kPa. [PU2004 Spring]
4. A vessel contains one kg of steam which contains $1/3$ liquid and $2/3$ vapor by volume. The temperature of the steam is $151.86^\circ C$. Find the quality, specific volume and specific enthalpy of the mixture. [PU 2006 Spring]

5. A vessel having a volume of $0.6 m^3$ contains 3 kg of liquid water and water vapor mixture in equilibrium at a pressure of 0.5 MPa. Calculate:
 - a) the volume and mass of liquid
 - b) the volume and mass of steam. [PU 2007 Fall/Spring]
6. A closed system containing dry saturated steam undergoes expansion according to the law $pv^n = c$ from an initial pressure of 10 bar to a final pressure of 2 bar. If the steam is finally wet with dryness fraction 0.85, evaluate the index of expansion and the work done.
7. A vessel contains wet steam at a pressure of 2 bar, 12% of whose mass is in liquid. Find the following: (a) temperature (b) dryness fraction (c) specific volume (d) specific enthalpy and (e) specific internal energy
8. A vessel of one cubic meter contains wet steam at a pressure of 2 bar, 12% of whose volume is liquid. Find the following: (a) temperature (b) dryness fraction (c) specific volume (d) specific enthalpy.

9. Saturated water vapour at $60^\circ C$ has its pressure decreased to increase the volume by 10% keeping the temperature constant. To what pressure should it be expanded?
10. Steam initially at 6 MPa and $450^\circ C$ is allowed to cool to $200^\circ C$ at constant volume. Find the final condition and heat transfer if the mass of steam is 0.75 kg.

11. Steam at 10 bar pressure and 0.9 dryness fraction is cooled at constant volume at 160°C .
What will be its final condition?
12. During a steady flow process, 5 kg of steam at 20 bar and 300°C loses 4150 kJ of heat at constant pressure. Determine the final condition of steam.
13. 0.1 kg steam at 5 bar, 0.9 dry expands to a pressure of 1 bar in a resisted process that follows the path $pv^{1.2} = \text{constant}$. Find work transfer, heat transfer and change in entropy.
14. Steam at 3 bar and 0.9 dry expands in a cylinder till the volume is four times that at the commencement. The law of expansion is $pv^{1.1} = c$. Determine the quality of steam after expansion.
15. A rigid cylinder contains steam at 10 bar and a temperature of 300°C . The steam is cooled until its pressure decreases to 1 bar. Find the final state and the heat transfer per kg of steam.

Pure Substance

Answers:

1. 0.0295
2. 0.0008 m^3 ; 0.7356 kg; 0.3992 m^3 ; 1.2644 kg
- 3.
4. 0.00579; 652.42 kJ/kg; $0.00326 \text{ m}^3/\text{kg}$
- 5.
6. 1.19; 230.74 kJ/kg
7. 120.23°C ; 0.88; $0.779 \text{ m}^3/\text{kg}$; 2442.4 kJ/kg; 2286.49 kJ/kg
8. 120.23°C ; 0.0087; $0.00876 \text{ m}^3/\text{kg}$; 523.84 kJ/kg
9. 18.9 kPa
10. 0.4038; -1423.35 kJ
11. 0.5706 dry
12. 0.68 dry
13. 198.75 kJ/kg; -163.55 kJ/kg
14. 0.834
15. 0.1517 dry; -2059.06 kJ/kg

Pure Substances

A two-phase liquid vapour mixture of H_2O is at 30 bar. If on heating at fixed volume, critical point is attained. Determine the quality at initial state.

solⁿ.
Here, At initial state

$$P_1 = 30 \text{ bar} = 3000 \text{ kPa}$$

At final state, critical point is obtained, from steam table.
sp. volume at final state (v_2) = $0.00319 \text{ m}^3/\text{kg}$

Then,

$$v_1 = v_2 \quad (\text{since the process is constant volume process})$$

$$v_1 = 0.00311 \text{ m}^3/\text{kg} = v$$

Now,

$v = v_f$ from steam table at 3000 kPa,

$$v_f = 0.001217 \text{ m}^3/\text{kg}$$

$$v_{fg} = 0.06544 \text{ m}^3/\text{kg}$$

Then,

$$v = v_f + X v_{fg}$$

$$0.00311 = 0.001217 + X \cdot 0.06544$$

$$1.893 \times 10^{-3} = X \cdot 0.06544$$

$$X = 0.0289$$

2. A vessel having a volume of 0.4 m^3 contains 2 kg of liquid water and water vapour mixture in equilibrium at a pressure of 0.5 MPa. Calculate

(a) the volume and mass of liquid

(b) the volume and mass of vapour

→ solⁿ.

Here,

$$\text{mass of the mixture (m)} = 2 \text{ kg}$$

$$\text{sat. pressure (P)} = 0.614 \text{ MPa} = 600 \text{ kPa}$$

$$\text{volume of the mixture (V)} = 0.4 \text{ m}^3$$

Now,

$$\text{we know that, sp. volume of mixture} = v_f = \frac{V}{m}$$

$$= \frac{0.4}{2}$$

$$\therefore v_f = 0.2 \text{ m}^3/\text{kg}$$

At 600 kPa

From steam table, we have

$$v_f = 0.001101 \quad v_{fg} = 0.3145 \quad v_g = 0.31560$$

Since, the mixture contains water and steam at equilibrium temperature, we have

$$v = v_f + x \cdot v_{fg}$$

$$\text{or, } 0.2 = 0.001101 + x \cdot 0.3145$$

$$\text{or, } x = \frac{0.198899}{0.3145}$$

$$x = 0.6324$$

Also,

$$x = \frac{m_g}{m}$$

$$\Rightarrow m_g = 2 \times 0.632$$

$$m_g = 1.264 \text{ kg}$$

$$\begin{aligned} \text{And, } m_f &= m - m_g \\ &= 2 - 1.264 \\ &= 0.735 \text{ kg} \end{aligned}$$

where m_f = mass of liquid
 m_g = mass of vapor
 x = dryness factor

sp. volume is defined as $v = \frac{V}{m}$

$$\Rightarrow V = v m$$

Then,

a) volume of liquid; $V_g = v_g m_g = 0.31560 \times 1.264$
 $= 0.398 \text{ m}^3$

b) volume of liquid and vapour $V_f = v_f m_f$
 $= 0.001101 \times 0.735$
 $= 0.000809 \text{ m}^3 //$

$\Rightarrow$ Hence, the mass of and volume of the liquid are 0.735 kg and 0.000809 m^3 and mass and volume of the vapour / steam are 0.735 kg and 0.398 m^3 respectively.

2) 5 kg of H_2O are contained in a closed rigid tank at an initial pressure of 2000 kPa and quality of 50%. Heat transfer occurs until the tank contains only saturated vapour. Determine the volume of the tank, in m^3 , and the final pressure, in kPa.

2. Solⁿ. -

A vessel contains one kg of steam which contains $\frac{1}{3}$ liquid and $\frac{2}{3}$ vapour by volume. The temperature of the steam is 151.86°C . Find the quality, sp. volume and sp. enthalpy of the mixture.

Let, V be the total volume of the mixture.
Then, volume of liquid (V_f) = $\frac{V}{3}$

$$\text{volume of vapour } (V_g) = \frac{2V}{3}$$

We have, total mass of mixture (m) = $m_f + m_g$

$$\left[\because v = \frac{V}{m} \right] \quad \frac{V}{v} = \frac{V_f}{v_f} + \frac{V_g}{v_g}$$

$$\frac{V}{v} = \frac{V}{3} + \frac{2V}{3} \cdot \frac{1}{v_g}$$

$$\text{or, } \frac{V}{v} = \frac{V}{3} \left[\frac{1}{v_f} + \frac{2}{v_g} \right]$$

$$\text{or, } 1 = \frac{1}{3} \left[\frac{1}{v_f} + \frac{2}{v_g} \right]$$

$$\Rightarrow v = \frac{3}{\left[\frac{1}{v_f} + \frac{2}{v_g} \right]} \quad \text{--- (i)}$$

At 151.86°C ,

from steam table, we have

$$v_f = 0.001098 \text{ m}^3/\text{kg}$$

$$v_{fg} = 0.3738 \text{ m}^3/\text{kg}$$

$$v_g = 0.3749 \text{ m}^3/\text{kg}$$

substituting these values in eqⁿ ①, we get

$$V = \frac{9}{\left[\frac{1}{0.001090} + \frac{1}{0.3749} \right]}$$

$$V = 3.26 \times 10^{-3} \text{ m}^3$$

Now,

$$\text{sp. volume } (v) = \frac{V}{m} = \frac{3.26 \times 10^{-3}}{1} = 3.26 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$\text{Then, } v = v_f + x v_{fg}$$

$$x = \frac{v - v_f}{v_{fg}}$$

$$= \frac{3.26 \times 10^{-3} - 0.001090}{0.3738}$$

$$x = 0.00579$$

Again,

from steam table,

$$h_f = 640.38 \text{ kJ/kg}$$

$$h_{fg} = 2108.2 \text{ kJ/kg}$$

$$h_g = 2748.6 \text{ kJ/kg}$$

Then,

$$h = h_f + x h_{fg}$$

$$= 640.38 + 0.00579 \times 2108.2$$

$$= 652.60 \text{ kJ/kg}$$

⑤ A vessel having a volume of 0.6 m^3 contains 3 kg of liquid water and water vapour mixture in equilibrium at a pressure of 0.5 MPa . Calculate:

- (a) the volume and mass of liquid
(b) the volume and mass of steam.

→ solⁿ:-

Here, volume of the mixture (V) = 0.6 m^3
mass of the mixture (m) = 3 kg
Equilibrium pressure (P) = $0.5 \text{ MPa} = 500 \text{ kPa}$

Then, sp. volume of mixture $v_s = \frac{V}{m}$

$$= \frac{0.6}{3}$$

$$= 0.2 \text{ m}^3/\text{kg}$$

Also, $X = m_g$

At 500 kPa

from steam table, we have,

$$v_f = 0.001093 \text{ m}^3/\text{kg}$$

$$v_{fg} = 0.3738 \text{ m}^3/\text{kg}$$

$$v_g = 0.3749 \text{ m}^3/\text{kg}$$

Then, $v_s = v_f + X v_{fg}$

$$\text{or, } 0.2 = 0.001093 + X \times 0.3738$$

$$\text{or, } X = \frac{0.198907}{0.3738}$$

$$X = 0.5321$$

$$\text{Also, } X = \frac{m_g}{m} \Rightarrow m_g = X m = 0.5321 \times 3 = 1.596 \text{ kg}$$

Now, $m = m_f + m_g$

$$m_f = 3 - 1.596$$

$$\therefore m_f = 1.403 \text{ kg}$$

But, $v = \frac{V}{m}$

Then, $v \Rightarrow V = v m$

Then,

$$\therefore V_f = v_f m_f = 0.001093 \times 1.493 = 0.00153 \text{ m}^3$$

$$\therefore V_g = v_g m_g = 0.3749 \times 1.596 = 0.598 \text{ m}^3$$

⑥ A closed system containing dry saturated steam undergoes expansion according to the law $p v^n = c$ for an initial pressure of 10 bar to a final pressure of 2 bar. If the steam is finally wet with dryness fraction 0.85, evaluate the index of expansion and the work done.

$\rightarrow$ Solⁿ:

Here, At initial state

$$p_1 = 10 \text{ BAR} = 1000 \text{ KPa}$$

At final state

$$p_2 = 2 \text{ BAR} = 200 \text{ KPa}$$

$$x = 0.85$$

At 200 KPa,

$$v_f = 0.001060 \text{ m}^3/\text{kg}$$

$$v_{fg} = 0.8848 \text{ m}^3/\text{kg}$$

$$v_g = 0.8859 \text{ m}^3/\text{kg}$$

Now,

$$v = v_f + x \cdot v_{fg}$$

$$= 0.001060 + 0.85 \times 0.8848$$

$$v = 0.753 \text{ m}^3/\text{kg}$$

$$\therefore v_2 = v = 0.753 \text{ m}^3/\text{kg}$$

At 1000 KPa

$$v_f = 0.001227 \text{ m}^3/\text{kg}$$

$$v_g = 0.1944 \text{ m}^3/\text{kg}$$

Here, system contains dry saturated steam initially so

$$v_1 = v_g = 0.1944 \text{ m}^3/\text{kg}$$

since, the process is polytropic one,

$$\text{So, } P_1 V_1^n = P_2 V_2^n$$

$$\text{or, } \frac{P_1}{P_2} = \left(\frac{V_2}{V_1} \right)^n$$

$$\text{or, } \ln \frac{P_1}{P_2} = n \ln \left(\frac{V_2}{V_1} \right)$$

$$\text{or, } \ln \left(\frac{1000}{200} \right) = n \ln \left(\frac{0.753}{0.1944} \right)$$

$$\text{or, } 1.60 = n \times 1.35$$

$$\therefore n = \frac{1.18}{1.19}$$

Hence,

$$W = \frac{P_2 V_2 - P_1 V_1}{1-n}$$

$$= \frac{200 \times 1000 \times 0.753 - 1000 \times 1000 \times 0.1944}{1 - 1.18/1.19}$$

$$= 243.33 \text{ kJ/kg} = 230.526.31 \text{ kJ/kg}$$

$$= 230.52 \text{ J/kg}$$

⑦ A vessel contains wet steam at a pressure of 2 BAR. 12% of whose mass is in liquid. Find the following

(a) Temperature

(b) dryness factor

(c) specific volume

(d) specific enthalpy

(e) specific internal energy

-2.01

Here,

$$P = 2 \text{ BAR} = 200 \text{ kPa}$$

% of mass of liquid = 12%

Then, % of vapour in the wet steam = $100 - 12$
= 88%

$$\begin{aligned}\text{Then, (b) } x &= \frac{m_g}{m} \\ &= \frac{88\% \text{ of } m}{m} \\ &= \frac{88}{100} \times m \\ &\quad m \\ &= 0.88\end{aligned}$$

At 200 kPa, $v_f = 0.001060 \text{ m}^3/\text{kg}$

$$v_{fg} = 0.8848 \text{ m}^3/\text{kg}$$

$$v_g = 0.8859 \text{ m}^3/\text{kg}$$

$$\begin{aligned}\text{(c) Then, } v &= v_f + x \cdot v_{fg} \\ &= 0.001060 + 0.88 \times 0.8848 \\ &= 0.7796 \text{ m}^3/\text{kg}\end{aligned}$$

(d) From steam table, $t = 120.24^\circ\text{C}$

$$\begin{aligned}\text{(d) From steam table, } h_f &= 504.80 \text{ kJ/kg} \\ h_{fg} &= 2201.7 \text{ kJ/kg} \\ h_g &= 2706.5 \text{ kJ/kg}\end{aligned}$$

$$\begin{aligned}\text{Then, } h &= h_f + x \cdot h_{fg} \\ &= 504.80 + 0.88 \times 2201.7 \\ &= 2442.29 \text{ kJ/kg}\end{aligned}$$

② from steam table, $u_f = 504.59 \text{ kJ/kg}$
 $u_{fg} = 2024.8 \text{ kJ/kg}$
 $u_g = 2529.4 \text{ kJ/kg}$

Then, $u = u_f + x \cdot u_{fg}$
 $= 504.59 + 0.88 \times 2024.8$
 $= 2286.414 \text{ kJ/kg}$

⑧ A vessel of one cubic meter contains wet steam at a pressure of 2 BAR, 12% of whose volume is liquid. find the following: (a) temperature, (b) dryness fraction (c) specific volume (d) specific enthalpy.
 → Solⁿ:-

volume of wet steam (V) = 1 m^3
 pressure (P) = 2 BAR = 200 KPA

(a) temperature: At 200 KPA
 from steam table, $t = 120.24^\circ\text{C}$

(b) dryness fraction

Here, volume of liquid (V_f) = 12% of V
 volume of vapour (V_g) = (100 - 12)% = 88% of V

Then, $V = V_f + V_g$
 $\text{or, } V = 12\% \text{ of } V + 88\% \text{ of } V$
 $\text{or, } 1 = \frac{12 \times V}{100} + \frac{88 \times V}{100}$

$m = m_f + m_g$
 $\frac{V}{v} = \frac{V_f}{v_f} + \frac{V_g}{v_g}$ ——— ①

- At 200 kPa

from steam tables $v_f = 0.001060 \text{ m}^3/\text{kg}$

$$v_{fg} = 0.8848 \text{ m}^3/\text{kg}$$

$$v_g = 0.8859 \text{ m}^3/\text{kg}$$

Then, in eqn (i),

$$\frac{1}{v} = \frac{12\% \text{ of } v}{v_f} + \frac{88\% \text{ of } v}{v_g}$$

$$\text{or, } \frac{1}{v} = \frac{\frac{12}{100} \times 1}{0.001060} + \frac{\frac{88}{100} \times 1}{0.8859}$$

$$\text{or, } \frac{1}{v} = 113.20 + 0.9923$$

$$\text{or, } \frac{1}{v} = 114.19$$

$$\therefore v = 0.00875 \text{ m}^3/\text{kg}$$

Now,

$$v = v_f + x \cdot v_{fg}$$

$$\text{or, } 0.00875 = 0.001060 + x \times 0.8849$$

$$\therefore x = 0.0087$$

$$\textcircled{c} \text{ sp. volume } (v) = 0.00875$$

$$\textcircled{d} \text{ sp. enthalpy; } h_f = 504.80 \text{ kJ/kg}$$

$$h_{fg} = 2201.7 \text{ kJ/kg}$$

$$\text{Then, } h = h_f + x \cdot h_{fg}$$

$$= 504.80 + 0.0087 \times 2201.7$$

$$= 523.95 \text{ kJ/kg}$$

1A

Q) Saturated water vapour at 60°C has its pressure decreased to increase the volume by 10% keeping the temperature constant. To what pressure should it be expanded?

solⁿ:-

$$t = 60^\circ\text{C}$$

From steam table,

$$\text{At } 60^\circ\text{C}, v_g = 7.6743 \text{ m}^3/\text{kg}$$

Then,

$$v' = v_g + 10\% \text{ of } v_g$$

$$= \frac{10}{100} \times v_g + v_g = 7.6743 + \frac{10}{100} \times 7.6743$$

$$= 8.44173 \text{ m}^3/\text{kg}$$

Superheated steam can be treated as the ideal gas.

$$\text{So, } \frac{p v}{T} = \text{constant}$$

Also, t is constant in this process.

$$p v = \text{const.}$$

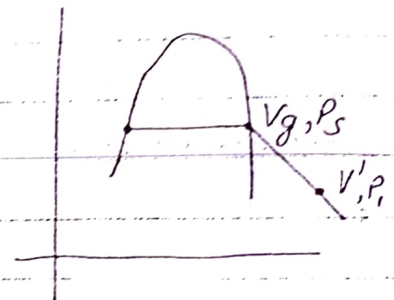
$$\text{a, } p_s v_g = p' v'$$

$$\text{a, } 14.932 \times 7.6743 = p' \times 8.44173$$

$$\text{a, } 113.6099 = p' \times 8.44173$$

$$\text{a, } 13.4571 = p'$$

$$\therefore p' = 13.46 \text{ kPa}$$



(10) Steam initially at 6 MPa and 450°C is allowed to cool at 200°C at constant volume. Find the final condition and heat transfer if the mass of steam is 0.75 kg.

→ Soln:-

At initial state

$$P_1 = 6 \text{ MPa} = 6000 \text{ kPa}$$

$$t_1 = 450^\circ\text{C}$$

At final state

$$t_2 = 200^\circ\text{C}$$

At 6000 kPa,

from steam table, $v_f = 0.001319 \text{ m}^3/\text{kg}$

$$v_{fg} = 0.03112 \text{ m}^3/\text{kg}$$

$$v_g = 0.03244 \text{ m}^3/\text{kg}$$

$$v_s = 0.05214 \text{ m}^3/\text{kg}$$

$$v_s = v_1 = 0.05214 \text{ m}^3/\text{kg}$$

Then,

$$t_s = 275.62^\circ\text{C}$$

since, $t_s < t_1$: so, the steam must be superheated
or:

Also,

$$v_1 = v_2 = 0.05214 \text{ m}^3/\text{kg} \text{ (since process is const. vol}^m\text{ process)}$$

At 200°C

from steam table, $v_f = 0.001156 \text{ m}^3/\text{kg}$

$$v_{fg} = 0.1261 \text{ m}^3/\text{kg}$$

$$v_g = 0.1273 \text{ m}^3/\text{kg}$$

$$\text{Also, } v = v_1 = v_2 = 0.05214 \text{ m}^3/\text{kg}$$

Then, $U = U_f + X \cdot U_{fg}$

$0.05214 \times 0 = 0.001156 + X \cdot 0.1261$

$\therefore X = 0.407$

$\therefore$ dryness factor = 0.407

Now, At 6000 kPa

$U_1 = 2989.1 \text{ kJ/kg}$

Then, for U_2

$U_2 = U_f + X \cdot U_{fg}$

At 200°C

$U_f = 850.58 \text{ kJ/kg}$

$U_{fg} = 1744.1 \text{ kJ/kg}$

Then,

$U_2 = 850.58 + 0.407 (1744.1)$
 $= 1555.19 \text{ kJ/kg}$

Now,

total heat transfer $\Rightarrow \delta Q = du + \delta w$

${}_1Q_2 = \Delta U + {}_1W_2$

${}_1Q_2 = \Delta U + 0$ [since process is isochoric]

${}_1Q_2 = U_2 - U_1$

$= 1555.19 - 2989.1$

$= -1433.9036 \text{ kJ/kg}$

Then, net work transfer = ${}_1Q_2 \times m$

$= -1433.9036 \times 0.75$

$= -1075.427 \text{ kJ}$

Q12) steam at 10 BAR pressure and 0.9 dryness fraction is cooled at const. volume at 160°C . What will be its final condition.

→ solⁿ:-

At initial condition,

$$P_1 = 10 \text{ BAR} = 1000 \text{ kPa}$$

$$x = 0.9$$

At final condition

$$t_2 = 160^\circ\text{C}$$

process is const volume process.

At 1000 kPa pressure,

from steam table,

$$v_f = 0.001127 \text{ m}^3/\text{kg}$$

$$v_{fg} = 0.1933 \text{ m}^3/\text{kg}$$

$$v_g = 0.1944 \text{ m}^3/\text{kg}$$

$$x = 0.9$$

$$\text{Then, } v = v_f + x \cdot v_{fg}$$

$$= 0.001127 + 0.9 \times 0.1933$$

$$= 0.175097 \text{ m}^3/\text{kg}$$

$$v = v_1 = v_2 = 0.175 \text{ m}^3/\text{kg} \quad \text{since the process is isochoric (isochoric means)}$$

At 160°C

from steam table,

$$v_f = 0.001102 \text{ m}^3/\text{kg}$$

$$v_{fg} = 0.3060 \text{ m}^3/\text{kg}$$

$$v_g = 0.3071 \text{ m}^3/\text{kg}$$

$$\text{Then, } v = v_f + x \cdot v_{fg}$$

$$\text{or, } 0.175 = 0.001102 + x \cdot 0.3060$$

$$\therefore x = 0.57$$

Also, here, $v_f < v < v_g$ so, the

⑫ During a steady flow process 5 kg of steam of 20 bar and 300°C and loses 4150 kJ of heat at constant pressure. Determine the final condition of steam.
 → solⁿ.

At initial condition

$$P_1 = 20 \text{ BAR} = 2000 \text{ kPa}$$

$$t_1 = 300^\circ\text{C}$$

process is iso-baric one

So,

at final condition

$$P_2 = P_1 = 2000 \text{ kPa}$$

and

$$\dot{Q} = -4150 \text{ kJ}$$

Then, At initial condition, at 2000 kPa, from steam table
 $t_s = 212.42^\circ\text{C}$, since $t_1 > t_s$ so, the steam is superheated one in initial condition.

$$\therefore h_1 = 3022.7 \text{ kJ/kg}$$

For a constant pressure process, net heat transfer is equal to change in enthalpy i.e.

$$\dot{Q} = \dot{H} \Rightarrow \dot{Q}_2 = h_2 - h_1$$

$$\text{or, } -4150 = h_2 - h_1 \quad h_2 - h_1$$

$$\text{or, } -4150 = h_2 - 3022.7$$

$$\therefore h_2 = -1127.3 \text{ kJ/kg}$$

At final condition, at 2000 kPa,

$$h_f = 908.69 \text{ kJ/kg} \quad h_{fg} = 1890 \text{ kJ/kg}$$

$$\text{Then, } h_2 = h_f + x h_{fg}$$

$$\text{or, } -1127.3 = 908.69 + x \times 1890$$

$$\therefore x =$$

solⁿ:

At initial condition:-

$$P_1 = 5 \text{ bar} = 5 \times 10^5 \text{ Pa} = 500 \text{ kPa}$$

$$x_g = 0.9$$

At 500 kPa, from steam table, $v_f = 0.001093 \text{ m}^3/\text{kg}$
 $v_{fg} = 0.3738 \text{ m}^3/\text{kg}$

$$\begin{aligned} v_1 &= v_f + x_g v_{fg} \\ &= 0.001093 + (0.9 \times 0.3738) \\ &= 0.337 \text{ m}^3/\text{kg} \end{aligned}$$

At final condition:

$$P_2 = 1 \text{ Bar} = 1 \times 10^5 \text{ Pa} = 10^2 \text{ kPa}$$

The process is an expansion process complying

$$p v^{1.1} = c$$

$$\alpha, P_1 v_1^{1.1} = P_2 v_2^{1.1}$$

$$\alpha, 5 \times 10^5 \times 0.337^{1.1} = 10^5 \times v_2^{1.1}$$

$$\alpha, 151134.25 = 10^5 \times v_2^{1.1}$$

$$\therefore v_2 = 1.45 \text{ m}^3/\text{kg}$$

After expansion, volume of steam

$$V_2 = v_2 \times m = 1.45 \times 0.1 = 0.145 \text{ m}^3$$

$$\text{Also, } V_1 = v_1 \times m = 0.337 \times 0.1 = 0.0337 \text{ m}^3$$

Then,

$$\text{work transfer } ({}_1W_2) = \frac{P_2 V_2 - P_1 V_1}{1-n}$$

$$= \frac{10^5 \times 1 \times 0.145 - 5 \times 10^5 \times 0.0337}{1-1.2}$$

$$= 11750 \text{ J}$$

$$= 11.75 \text{ kJ}$$

Again, at 500 kPa, $u_f = 639.84 \text{ kJ/kg}$
 $u_{fg} = 1921.4 \text{ kJ/kg}$

$$\begin{aligned}
 u_1 &= u_f + x \cdot u_{fg} \\
 &= 639.84 + (0.9 \times 1922.4) \\
 &= 2369.1 \text{ kJ/kg}
 \end{aligned}$$

$$\therefore U_1 = u_1 \times m = 2369.1 \times 0.1 = 236.91 \text{ kJ}$$

Similarly, for final condition,

$$\begin{aligned}
 \text{At } 100 \text{ kPa, } v_f &= 0.001043 \text{ m}^3/\text{kg}, h_f = 417.51 \text{ kJ/kg} \\
 v_{fg} &= 1.6933 \text{ m}^3/\text{kg}, h_{fg} = 2257.6 \text{ kJ/kg}
 \end{aligned}$$

$$\begin{aligned}
 v_2 &= v_f + x \cdot v_{fg} \\
 \text{or, } 1.45 &= 0.001043 + x \times 1.6933 \\
 \therefore x &= 0.85
 \end{aligned}$$

$$\text{Then, } h_2 = u_f + x \cdot u_{fg} \quad u_{fg} = 2088.3 \text{ kJ/kg}$$

$$\begin{aligned}
 u_2 &= u_f + x \cdot u_{fg} \\
 &= 417.41 + (0.85 \times 2088.3) \\
 &= 2204.36
 \end{aligned}$$

$$\therefore U_2 = u_2 \times m = 2204.36 \times 0.1 = 220.43 \text{ kJ}$$

Then, from 1st law of thermodynamics, we have,

$$dq = \Delta u + w_2$$

$$\text{or, } dq = u_2 - u_1 + 11.75$$

$$\text{or, } dq = 220.43 - 236.91 + 11.75$$

$$\therefore dq = -4.73 \text{ kJ}$$

$$\therefore dq = \frac{dq}{m} = \frac{-4.73}{0.1} = -47.3 \text{ kJ/kg}$$

Q19. solⁿ:-

Here,

At initial condition;

$$P_1 = 3 \text{ bar} = 3 \times 10^5 \text{ Pa} = 300 \text{ kPa}$$

$$x_1 = 0.9$$

At 300 kPa, from steam table

$$v_f = 0.001073 \text{ m}^3/\text{kg}$$

$$v_{fg} = 0.6048 \text{ m}^3/\text{kg}$$

Then,

$$\begin{aligned} v_1 &= v_f + x \cdot v_{fg} \\ &= 0.001073 + (0.9 \times 0.6048) \\ &= 0.5453 \text{ m}^3/\text{kg} \end{aligned}$$

At final condition,

$$v_2 = 4v_1$$

The process follows the expansion law

$$p v^{1.1} = C$$

$$\text{or, } p_1 v_1^{1.1} = p_2 v_2^{1.1}$$

$$\text{or, } 3 \times 10^5 \times \left(\frac{v_1}{v_2} \right)^{1.1} = p_2$$

$$\text{or, } p_2 = 3 \times 10^5 \times \left(\frac{v_1}{4v_1} \right)^{1.1}$$

$$p_2 = 65291.29 \text{ Pa}$$

$$p_2 = 65.29 \text{ kPa}$$

$$v_2 = 4v_1 = 4 \times 0.5453 = 2.1812 \text{ m}^3/\text{kg}$$

At 65.29 kPa, the properties of steam can not be directly obtained from steam table. so, we have to determine the properties by interpolating as follows:

pressure (kPa)	v_f (m^3/kg)	v_{fg} (m^3/kg)
60	0.001033	2.7314
65.29	?	?
70	0.001036	2.7644

Interpolating for v_f , we get,

$$v_f - 0.001033 = \frac{0.001036 - 0.001033}{70 - 60} (65.29 - 60)$$

$$v_f - 0.001033 = 1.587 \times 10^{-6}$$

$$v_f = 1.03 \times 10^{-3} \text{ m}^3/\text{kg}$$

Interpolating for v_{fg} , we get,

$$v_{fg} - 2.7314 = \frac{2.7644 - 2.7314}{70 - 60} (65.29 - 60)$$

$$v_{fg} - 2.7314 = 0.1941$$

$$v_{fg} = 2.537 \text{ m}^3/\text{kg}$$

so, at 65.29 kPa,

$$v_2 = v_f + x \cdot v_{fg}$$

$$2.1812 = 1.03 \times 10^{-3} + x \cdot 2.537$$

$$x = 0.859 //$$

Q5) Solⁿ:-

Here, At initial condition,

$$P_1 = 10 \text{ bar} = 10 \times 10^5 \text{ Pa} = 10 \times 10^2 = 1000 \text{ kPa}$$

$$t_1 = 300^\circ\text{C}$$

At final condition,

$$P_2 = 1 \text{ bar} = 10^5 \text{ Pa} = 100 \text{ kPa}$$

At 1000 kPa, from steam table,

$$t_s = 179.92^\circ\text{C}$$

Since, $t_1 > t_s$. Hence, the steam is superheated one.

$$\therefore v_1 = 0.2579 \text{ m}^3/\text{kg}$$

$$h_1 = 3050.6 \text{ kJ/kg}$$

$$u_1 = 2792.7 \text{ kJ/kg}$$

$$s_1 = 7.1219 \text{ kJ/kg}\cdot\text{K}$$

Prabin chaudhary

NAST

BE Computer II
(2022 spring)